

PROBABILITY DENSITY FUNCTION

1. Given the following probability function find the value 'a'

$$f(x) = \begin{cases} ax & 0 \leq x \leq 1 \\ a & 1 \leq x \leq 2 \\ -ax + 3a & 2 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

Also compute i) $P(\frac{1}{2} \leq x \leq \frac{3}{2})$ ii) $P(x \leq 2)$ iii) $P(\frac{3}{2} \leq x \leq 3)$

Given that $f(x)$ is a Probability density function.

$$\therefore \int_{-\infty}^{\infty} f(x) dx = 1$$

$$\Rightarrow \int_0^3 f(x) dx = 1$$

$$\Rightarrow \int_0^1 f(x) dx + \int_1^2 f(x) dx + \int_2^3 f(x) dx = 1$$

$$\Rightarrow \int_0^1 ax dx + \int_1^2 a dx + \int_2^3 (-ax + 3a) dx = 1$$

$$\Rightarrow \left(\frac{ax^2}{2} \right)_0^1 + (ax)_1^2 - \left(\frac{ax^2}{2} \right)_2^3 + (3ax)_2^3 = 1$$

$$\Rightarrow \frac{a}{2} + a - \frac{5a}{2} + 3a = 1$$

$$\Rightarrow a = \frac{1}{2}$$

$$\text{i) } P\left(\frac{1}{2} \leq x \leq \frac{3}{2}\right) = \int_{1/2}^{3/2} f(x) dx = \int_{1/2}^1 f(x) dx + \int_1^{3/2} f(x) dx$$

$$= \int_{1/2}^1 ax dx + \int_1^{3/2} a dx$$

$$= \left(\frac{ax^2}{2} \right)_{1/2}^1 + (ax)_1^{3/2} = 7/16$$

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ii)

$$P(x \leq 2) = \int_0^2 f(x)dx = \int_0^1 f(x)dx + \int_1^2 f(x)dx = \int_0^1 axdx + \int_1^2 adx \\ = 3/4$$

iii)

$$P\left(\frac{3}{2} \leq x \leq 3\right) = \int_{3/2}^3 f(x)dx \\ = \int_{3/2}^2 f(x)dx + \int_2^3 f(x)dx \\ = \int_{3/2}^2 adx + \int_2^3 (-ax + 3a)dx = 1/2$$

Properties for probability density function

The probability density function $f(x)$ of a continuous random variable X satisfies the following properties.

1) $f(x) \geq 0$, for all x

A random variable X may assume negative values from $-\infty$ to 0 . But $f(x)$ cannot take negative values as it defines probability measure.

(2) $\int_{-\infty}^{\infty} f(x)dx = 1$

3) $P(x_1 \leq x \leq x_2) = \int_{x_1}^{x_2} f(x)dx$

4) $P(x = a) = \int_a^a f(x)dx = 0$

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Relation between probability distribution function F(x) and probability density function $f(x)$

The probability density function is defined as the derivative of the distribution function F(x) as

$$f(x) = \frac{d}{dx} F(x)$$

Cumulative distribution function

Cumulative distribution functions (or) distribution function of random variable X is defined as

$$F(x) = \int_{-\infty}^x f(x)dx \quad (\text{or}) \quad f(x) = F'(x)$$

Problems:

1. If a random variable X has a cumulative distribution function F given by

$$F(x) = \begin{cases} 0 & \text{for } x \leq 0 \\ c(1 - e^{-x}) & \text{for } x > 0 \end{cases}$$

Find the (i) p.d.f of f(x) (ii) value of c.

Since $f(x) = F'(x)$

$$f(x) = F'(x) = \begin{cases} 0 & \text{for } x \leq 0 \\ ce^{-x} & \text{for } x > 0 \end{cases}$$

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

$$\text{But } \therefore c \int_0^{\infty} e^{-x} dx = c(-e^{-x})_0^{\infty} = c(e^{-\infty} + e^{-0}) = 1$$

$$\Rightarrow c = 1$$

$$\therefore f(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ e^{-x} & \text{if } x > 0 \end{cases}$$

2. X is a continuous random variable with pdf

$$f(x) = \begin{cases} kx, & 0 \leq x \leq 2 \\ 2k, & 2 \leq x \leq 4 \\ (6k - kx), & 4 \leq x \leq 6 \\ 0, & \text{otherwise} \end{cases}$$

Find the value of 'k' and also find the CDF for f(x).

We know that, the total probability is 1 if f(x) is a pdf.

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

i.e.,

$$\Rightarrow \int_0^2 kx dx + \int_2^4 2k dx + \int_4^6 (6k - kx) dx = 1 = 1/8$$

Cumulative distribution function:

In the interval $0 \leq x \leq 2$

$$F(x) = P(X \leq x) = \int_0^x f(x) dx$$

$$= \int_0^x kx dx$$

$$= \left(\frac{1}{2} x^2 \right)_0^x = \frac{x^2}{2}$$

In the interval $2 \leq x \leq 4$

$$F(x) = P(X \leq x) = \int_0^x f(x) dx$$

$$= \int_0^2 kx dx + \int_2^x 2k dx$$

$$= \left(\frac{kx^2}{2} \right)_0^2 + (2kx)_2^x$$

$$= 2k + 2k(x - 2)$$

$$= \frac{2}{2} + \frac{2}{2}(x - 2)$$

$$= \frac{x}{2} - \frac{1}{2}$$

In the interval $4 \leq x \leq 6$

$$\begin{aligned} F(x) &= P(X \leq x) = \int_0^x f(x) dx \\ &= \int_0^2 kx dx + \int_2^4 2k dx + \int_4^x (6k - kx) dx \\ &= \left(\frac{kx^2}{2} \right)_0^2 + (2kx)_2^4 + \left(6kx - k \frac{x^2}{2} \right)_4^x \\ &= 2k + 4k + 6k(x - 4) - \frac{k}{2}(x^2 - 16) \\ &= -\frac{x^2}{16} + \frac{3x}{4} - \frac{5}{4} \end{aligned}$$

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